Phase-III
Proposed Solution

Date	05 March 2026
Team ID	Team 154
Project Name	Insurance Fraud Detection using ML
Maximum Marks	4 Marks

Proposed Solution:

S.No.	Parameter	Description
1.	Problem Statement	The insurance industry suffers billions in losses annually due to fraudulent claims, which are difficult and slow to detect manually.
2.	Idea / Solution Description	An end-to-end Machine Learning system that uses 12 different classification models to predict whether a claim is "Fraudulent" or "Legal" via a Flask web interface.
3.	Novelty / Uniqueness	Unlike single-algorithm approaches, this system evaluates 12 state-of-the-art models (XGBoost, CatBoost, etc.) to ensure the highest mathematical accuracy for the specific dataset.
4.	Social Impact / Customer Satisfaction	Automating fraud detection reduces financial drain on the industry and allows honest policyholders to receive their claim payouts significantly faster.
5.	Business Model	Revenue Model: Subscription-based or per-claim processing fee model for insurance firms. It generates value by direct cost savings from prevented fraudulent payouts.
6.	Scalability of the Solution	The modular 3-tier architecture (Web UI, Flask Backend, ML Engine) allows the system to be scaled horizontally across cloud servers to handle thousands of claims simultaneously.